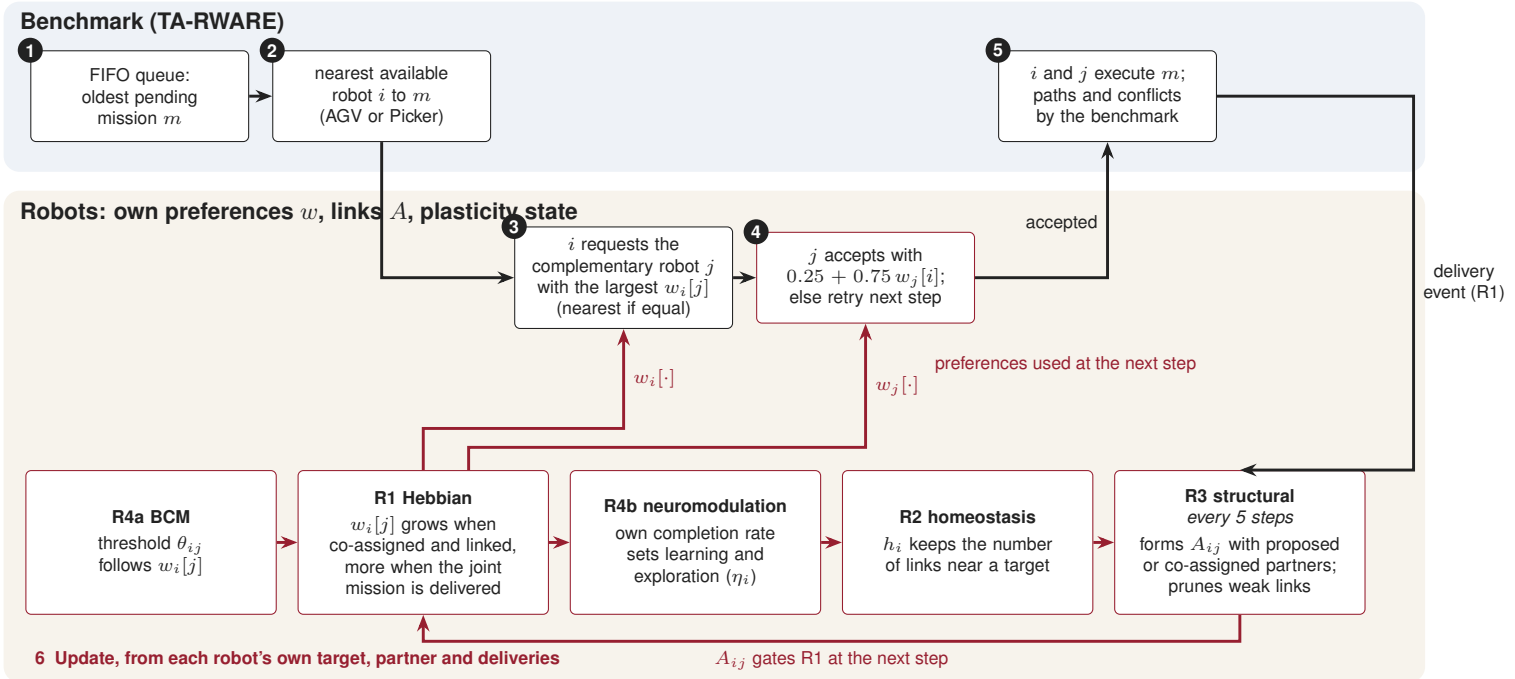

Benchmark (TA-RWARE)
1
FIFO queue: oldest pending mission m
2
nearest available robot i to m (AGV or Picker)
5
i and j execute m; paths and conflicts by the benchmark
Robots: own preferences w, links A, plasticity state
3
i requests the complementary robot j with the largest $w_i[j]$ (nearest if equal)
4
j accepts with $0.25 + 0.75\, w_j[i]$; else retry next step
accepted
delivery event (R1)
$w_i[\cdot]$
$w_j[\cdot]$
preferences used at the next step
R4a BCM
threshold θ_{ij} follows $w_i[j]$
R1 Hebbian
$w_i[j]$ grows when co-assigned and linked, more when the joint mission is delivered
R4b neuromodulation
own completion rate sets learning and exploration (η_i)
R2 homeostasis
h_i keeps the number of links near a target
R3 structural
every 5 steps
forms A_{ij} with proposed or co-assigned partners; prunes weak links
6 Update, from each robot's own target, partner and deliveries
A_{ij} gates R1 at the next step